

TIC 294976780

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 0.958619 d
Epoch (t0) 1543.9500 BTJD
Duration 0.1200 d
Depth 4.900 % (48,995 ppm)
SNR 133.1

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 48 (gate ≥ 3)
Robust depth ratio 1.05 (gate ≥ 0.50)
Robust depth sigma 59.0 (gate ≥ 7.0)
Odd/even z-score 1.53 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 9 light-curve dip in this sector: 59.3 sigma t0 refined by 0.0050 phase
Difference-image peak SNR 16.97 (gate ≥ 5.0) Compactness 1.00 (gate ≥ 0.50)
Dimming centroid offset 0.10 px (2 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 294976803
0.84 px away, dTmag +2.96 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)

